

APWEN · OYO STATE BRANCH

Building Ethical Engineers

in the Age of Artificial Intelligence

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WHERE THE PRESSURE REALLY COMES FROM

We were taught pressure has three variables.

Boyle's Law

Volume

Charles' Law

Temperature

Pressure Law

Molecules

Real-world engineering pressure is far more complex than that.

THE FOUR PLAYERS

**Between them, something
is always created.**

Clients

Engineering Firms

Artisans

Regulators (COREN)

It is called ethical pressure.

THE TRIPLE SQUEEZE

Time. Money. Scope.

Time

“We need it by Friday.”

Deadlines move. Standards
get skipped.

Money

“The other firm will do it
for less.” Fees pressed
down until honest work is
priced out.

Scope

“Just one more thing...” A
reasonable job quietly
becomes an impossible
one.

People beg and bargain until the impossible gives way. And what gives way with it is the standard.

WHY THIS ROOM MATTERS

Women engineers see what others overlook.



Precision of observation



Attention to the vulnerable



Planning for consequences

OUR CODE HAS ALWAYS BEEN CLEAR

What COREN frowns at



Unlicensed practice



Professional negligence



Falsified results



Contract corruption



Project abandonment



Inflated qualifications

Is technology good—
or evil?

THE ANSWER

Neither.

The morality is not in the tool.
It is in the application.

A knife. A drone. An algorithm. The tool does not choose — people choose.

THE BURDEN OF THE BUILDER

The Double Duty

1

What you think it is

The design. The intention. The brief.

2

How it can be used

By hands you will never meet,
in conditions you did not plan for.

THE ENGINEER'S HIGHEST CRAFT

Design out the cheating.



A pump that cannot be re-wired



A meter that cannot be bypassed



A record that cannot be back-dated

Make dishonesty impossible — by design.

**Money without work
is the mood of our age.**

Engineering has no shortcut.

WHY IT MATTERS

A corner cut is not a small sin.

**A bridge that
falls**

**A building that
collapses**

**A device that
kills**

LOOK AROUND US

How systems are cheated



Energy theft

Meters bypassed, power stolen



Audit fraud

Books cooked to look clean



E-voting attacks

Counting machines manipulated



Substandard goods

Faulty products, no maintenance

CLOSEST TO HOME

The construction site



Concrete watered down to pour easy



Grade 60 iron swapped for grade 40



Test results quietly back-dated

And when the building falls — where was the engineer?

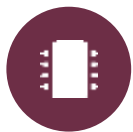
THE NEW ADVANTAGE

AI gives the engineer eyes everywhere.



Drones

Daily overflights, AI-read imagery



Sensors

Curing, density, integrity — live



Audit trails

Every delivery tied to a test



AI: a tireless conscience over our institutions.

It watches what no human has the patience to watch.

BUT WAIT

AI itself can be unethical.



Biased data

Prejudice wearing the mask of objectivity



Data ownership

Who owns it? Who consented? Who profits?



Digital divide

Designs that lock the excluded out

THE HEART OF THE MATTER

Two ethics, not one.

I

The profession

Competence. Honesty.
Safety. Integrity.

II

The technology

Fair data. Just access.
No one left behind.

DRAW THE LINE PLAINLY

Ethical / Not ethical



Public safety first



“I do not yet know”



Honest data and reports



Designing against misuse



Releasing what you know is dangerous



Pretending to unearned mastery



Fabricating figures



“I only built it”

BUILDING THEM

Where ethical engineers come from



Education

Ethics woven into the
craft



Society

Reward honest labour



Mentorship

Ethics is caught, not
taught



Forums

Name the problem out
loud

STUDENTS ARE USING AI. NOW WHAT?

Don't fight AI — teach them to lead it.

1

Ask what AI alone cannot answer

Local judgement. Context. Defence under pressure.

2

Make them audit AI outputs

Ten designs: which fail our codes, our soils?

3

Train them to detect the fake

In images, data and reports

THE HARDEST OBJECTION

Make integrity profitable.

Fair pricing for honest work. Standards with teeth.

Markets that prize the properly made.

*When doing it right is also the way to get paid,
ethics stops being a sacrifice — and becomes good sense.*

The bridge does not ask
whether you were in a hurry.

**It only asks whether
you built it right.**

Thank you. · Dr. Olusola Sayeed Ayoola